Ge 158 Lecture 7: Scattering from Nonplanar Surfaces

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Topics covered

- Radar equation
- Simple analytical models for scattering from rough surfaces
- Geometric properties of rough surfaces
- Scattering from undulating surfaces

1 Models for microwave backscattering from rough surfaces

1.1 Motivation

Motivation for focusing on microwave scattering:

- reinforce and expand on basic concepts discussed already;
- useful for SAR observations;
- sets up discussions of volume scattering and radiometry, which we'll discuss in the next few classes.

1.2 The radar equation

- Let's build more intuition for how we actually measure and quantify scattering.
- This is a heuristic model known as the "radar equation." Key components:
  1. transmit power from the antenna;
  2. radiated power from the target ("scatterer");
  3. effects of geometry, particularly spreading.

These components are sketched in Figure 1.

1.2.1 Spreading

$\hookrightarrow$ Spreading comes from the fact that radiated energy spreads as a sphere

$\Rightarrow$ the power density (W/m^2) drops off as $1/R^2$ ($4\pi R^2$ is the surface area of a sphere with radius R).

$\hookrightarrow$ This is consistent with the plane-wave assumption that we have been making, because the curvature of the sphere is negligible in the "far field," where $R \gg \lambda$ (wavelength) (and, for a source of largest dimension d_{ant}, $R \geq 2d_{\text{ant}}^2/\lambda$).

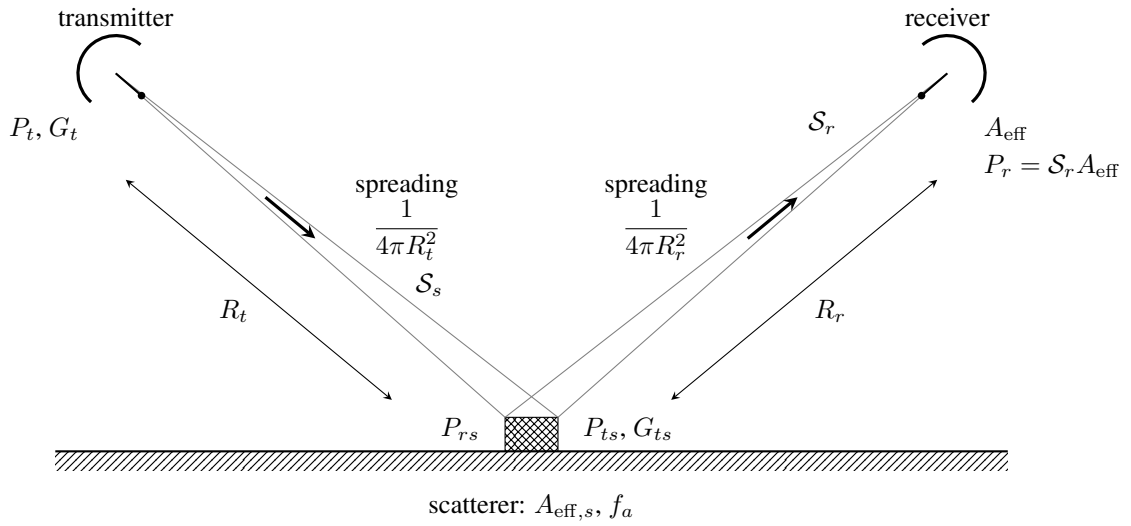


Figure 1: Components of the radar equation (Section 1.2). The transmitter radiates power P_t with gain G_t; the power density spreads as $1/(4\pi R_t^2)$, giving $\mathcal{S}_s$ at the scatterer. The scatterer intercepts $P_{rs} = \mathcal{S}_s A_{\text{eff},s}$, absorbs a fraction f_a, and reradiates $P_{ts} = P_{rs}(1 - f_a)$ with gain G_{ts} toward the receiver; after spreading as $1/(4\pi R_r^2)$, the power density $\mathcal{S}_r$ at the receiver gives the received power $P_r = \mathcal{S}_r A_{\text{eff}}$.

1.2.2 From the antenna to the scatterer

- Let's start from the source.
- Consider an antenna with transmit power P_t and gain G_t. This has power per unit solid angle at the antenna (in the direction of the scatterer) $= P_t G_t/(4\pi)$; $P_t G_t$ is the power that an isotropic antenna would have to transmit to produce the same power density.
- At the scatterer, a distance R_t from the antenna, we need to take into account the reduction of power density due to spreading. So, the power density at the scatterer (magnitude of the Poynting vector) is

$$\mathcal{S}_s = [P_t G_t] \left(\frac{1}{4\pi R_t^2} \right), \qquad \frac{\text{power}}{\text{area}} = \left[\begin{array}{c} \text{power transmitted} \\ \text{from antenna} \end{array} \right] \left(\begin{array}{c} \text{spreading} \\ \text{losses} \end{array} \right). \tag{1}$$

- The total power received by the scatterer, P_{rs}, must account for the effective receiving area of the scatterer, $A_{\text{eff},s}$:

$$\Rightarrow \quad P_{rs} = \mathcal{S}_s A_{\text{eff},s}. \tag{2}$$

★ Note: $A_{\text{eff},s}$ is <u>not</u> the area illuminated by the incident beam but rather an unknown property of the scatterer.

1.2.3 From the scatterer to the receiver

- Some power will be absorbed by the scatterer; the rest will be radiated in various directions. The power radiated by the scatterer is

$$P_{ts} = P_{rs}\,(1 - f_a)\,, \tag{3}$$

where $f_a \equiv$ fraction of power absorbed by the scatterer.

- Incident radiation will induce currents (free currents and bound currents due to conductivity and electric displacement → i.e., the non-magnetic terms in Ampère's law)

  ⇒ we can treat the scatterer as an antenna:

$$\mathcal{S}_r = \frac{P_{ts} G_{ts}}{4\pi R_r^2}, \tag{4}$$

  where

  * $G_{ts} \equiv$ gain of the scatterer in the direction of the receiver;
  * $R_r \equiv$ distance from the scatterer to the receiver.

- Finally, the power entering the receiver is a function of the effective area of the aperture, A_{eff} (not necessarily the physical size of the aperture):

$$P_r = \mathcal{S}_r A_{\text{eff}}. \tag{5}$$

1.2.4 Radar cross section

- Putting it all together, we can write an expression for the received power:

$$P_r = \underbrace{\left(\frac{P_t G_t A_{\text{eff}}}{(4\pi)^2 R_t^2 R_r^2} \right)}_{\text{knowns}} \underbrace{\left[A_{\text{eff},s} \left(1 - f_a\right) G_{ts} \right]}_{\text{unknowns}}. \tag{6}$$

- In general, the bracketed term is unknown and is a term of interest. Let us define it as

$$\sigma = A_{\text{eff},s} \left(1 - f_a\right) G_{ts} \qquad [\text{units: m}^2] \tag{7}$$

  and call it the *radar (scattering) cross section*, as it has units of m^2. (Note: in this class, I have tried to use variables in a way that is consistent with usage in RS, and here we have encountered one of the unfortunate inconsistencies in conventional variables, with σ now being used to denote something other than electrical conductivity.)

- More commonly, we normalize by the nominal area of illumination to get the normalized radar cross section ("sigma-naught"):

$$\sigma^0 = \left\langle \frac{\sigma_{(n)}}{A_{(n)}} \right\rangle_n, \tag{8}$$

  where $\sigma_{(n)} \equiv$ radar cross section for an individual scatterer, $A_{(n)} \equiv$ horizontal (ground-projected) area illuminated around scatterer n, and $\langle \cdot \rangle$ denotes the spatial average. The area is "nominal" because it is set by the radar geometry, not by the scatterer (cf. $A_{\text{eff},s}$). For a uniform distributed target of horizontal area A, Eq. (8) reduces to $\sigma^0 = \sigma / A$.

1.2.5 Accounting for polarization

- Up to this point, we have ignored wave polarization for simplicity. We can account for polarization by noting that the transmitted and received powers are polarization dependent:

  – $P_p^r \equiv p$-polarized received power (horizontal or vertical);
  – $P_q^t \equiv q$-polarized transmit power (h or v).

$$P_p^r = \frac{\lambda^2}{(4\pi)^3} \int_{\substack{\text{area} \\ \text{illuminated}}} \frac{P_q^t G^2}{R^4} \sigma_{pq}^0 \, dA, \tag{9}$$

where we have assumed

$$G_r = G_t = G \qquad \text{and} \qquad R_t = R_r = R \tag{10}$$

and applied the identity (for a lossless antenna)

$$A_{\text{eff}} = \frac{\lambda^2 G_r}{4\pi}. \tag{11}$$

- For intuition, we can note that

$$\sigma_{pq}^0 = \frac{4\pi R^2}{A} \frac{|E_p^s|^2}{|E_q^i|^2} = \frac{4\pi R^2}{A} \Gamma_{pq}, \tag{12}$$

where E_p^s is the scattered electric field (measured in the far field of the scatterer), E_q^i is the incident electric field, and Γ_{pq} is the reflectance, a generalization of the reflectance we derived before for planar surfaces (Lecture 5). Unlike the reflectivity $\Gamma = |\rho|^2$ of an infinite planar interface (Lectures 4–5), Γ_{pq} depends on the range R at which the scattered field is measured: the field scattered by the finite illuminated area A spreads spherically, so $|E_p^s|^2 \propto 1/R^2$. The factor $4\pi R^2/A$ in Eq. (12) removes this dependence, which makes σ_{pq}^0 a property of the surface alone.

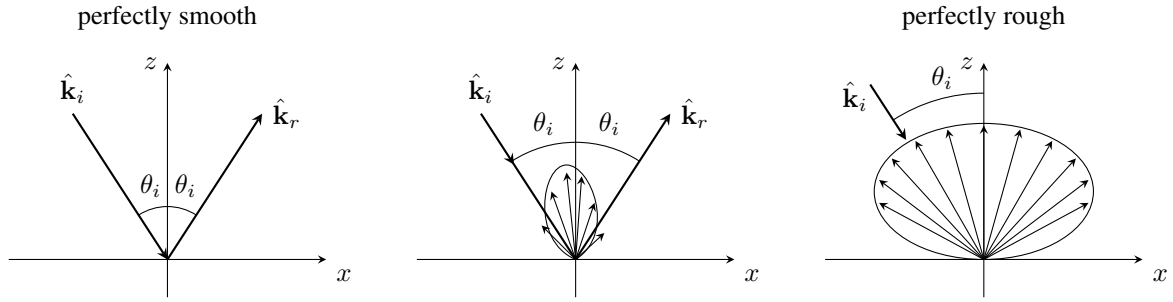
2 Scattering from nonplanar surfaces

2.1 Simple analytical models

- Now that we have a basic sense of how we often quantify the scattering of EM waves, let's consider some simple models for surfaces.
- We'll focus on simple analytical models.
- Simple analytical models only exist for a few special cases that occur when the spatial dimensions of the surface roughness are much smaller or much larger than the EM wavelength.
  - ↪ For the intermediate case, when roughness $\sim$ EM wavelength, very complex models are necessary. We won't go over these in class because they offer little additional insight. But one such model, known as the Integral Equation Model (IEM), is discussed in Ulaby and Long (2014), Chapter 10, and in the primary literature, Fung et al. (1992, 2002), for those who are interested.

2.2 Basic concepts of surface scattering

- For surface scattering, we are concerned with
  - the total power reflected by the surface (reflectivity);
  - the direction of scattering.
- Scattering from an area is simply (re)radiation from the area $\Rightarrow$ many concepts discussed here are adopted from antenna theory (which we will not discuss in detail, but those interested in learning more can consult Ulaby and Long (2014), Ch. 3).
- Key to understanding the direction of scatter is the surface roughness (Figure 2).



"specular" or coherent scattering

· reflected wave has uniform phase front ⇒ coherent

· all reflected power follows a single ray path

· combination of coherent and incoherent scatter

· natural surfaces typically fall in this category

"diffuse" or incoherent scatter

· a.k.a. Lambertian reflection

· radiance (apparent brightness) is the same in all directions

Figure 2: Scattering patterns for a plane wave incident at angle θ_i on a perfectly smooth surface (left), a surface of intermediate roughness (middle), and a perfectly rough surface (right).

2.3 Characterizing surface roughness

- How do we define surface roughness? Two length scales:
  1. RMS height h;
  2. correlation length ℓ.

2.3.1 Definitions

- Let us define the local surface height as a 2D function $\xi(x, y)$.
- Assume ξ has a Gaussian distribution with zero mean.
- Therefore, the RMS height is given by

$$h^2 = \left\langle \xi^2(x, y) \right\rangle . \tag{13}$$

- But we need more information, specifically how the height at one point relates to the height at another point

  ⇒ auto-correlation function (for a statistically isotropic surface)

$$\mathcal{C}(r) = \frac{\langle \xi(x, y)\, \xi(x', y') \rangle}{h^2}, \tag{14}$$

  where $r = \sqrt{(x - x')^2 + (y - y')^2}$ is the distance between the points.

2.3.2 Correlation functions

- Two correlation functions are commonly assumed:
  - Gaussian:

$$\mathcal{C}(r) = e^{-r^2/\ell^2}; \tag{15}$$

– Exponential:

$$\mathcal{C}(r) = e^{-|r|/\ell}. \tag{16}$$

$\Rightarrow$ This makes clear that ℓ is the distance over which the correlation decreases to $1/e$.

2.3.3 Where these correlation functions arise

- The key difference is the behavior near $r = 0$:
  - Gaussian: $\mathcal{C}(r) \approx 1 - r^2/\ell^2$ (flat top) $\rightarrow$ surface is smooth at scales $\ll \ell$; finite rms slope ($\sqrt{2}\,h/\ell$ in each direction); a single dominant roughness scale, since its roughness spectrum W (Eq. (19), below) falls off exponentially for $k\ell \gtrsim 2$.
  - Exponential: $\mathcal{C}(r) \approx 1 - r/\ell$ (cusp) $\rightarrow$ surface stays rough down to the smallest scales; rms slope is infinite; W (Eq. (20), below) falls off only as k^{-3} for $k\ell \gg 1$ (roughness at many scales).
- Exponential — natural surfaces:
  - Bare soils and agricultural/pasture fields: measured height profiles usually fit an exponential better than a Gaussian (e.g., about three-quarters of 117 profiles from fields in Oklahoma were close to exponential).
  - Lava flows (Mojave Desert): power-law roughness spectra, $W \propto k^{-2}$ to k^{-3}, similar to the tail of the exponential's spectrum; rough at every scale, like topography in general.
  - Snow and other granular, two-phase media (ice grains and air, soil grains and pores): along any line through the medium, crossings between phases occur at random, which gives an exponential correlation; microwave snow models describe snow microstructure by an "exponential correlation length" (relevant for volume scattering).
- Gaussian — surfaces (or fields) smoothed by some process:
  - An initially uncorrelated (white-noise) random field smoothed by diffusion has a Gaussian correlation function, with ℓ growing as the square root of time (e.g., a temperature or concentration field, or topography smoothed by soil creep); the smoothing removes the small scales.
  - Measured profiles smoothed by the instrument (finite laser spot or pin spacing) also lose their small scales and look more Gaussian near $r = 0$.
  - In practice, the Gaussian is mostly an analytical convenience; it is rarely the best fit to natural microtopography, but it is needed for models that require finite slopes (e.g., the geometric optics result, Eq. (21)).
- The same pair shows up in time, not space, in Lecture 11 (spectral line shapes):
  - Pressure broadening: collisions damp the oscillator, so its correlation in time has an exponential envelope $\Rightarrow$ Lorentz line shape.
  - Doppler broadening: Maxwellian (Gaussian) distribution of molecular velocities $\Rightarrow$ Gaussian correlation in time $\Rightarrow$ Gaussian line shape.
  - In both cases the line shape is the Fourier transform of the correlation function (Wiener–Khinchin), just as W is the transform of $\mathcal{C}$ below (in 1D, the exponential transforms to a Lorentzian; the 2D isotropic transform in Eq. (20) has a different form).

2.3.4 *Roughness spectrum*

- It is more common to represent the surface in terms of the roughness spectrum, the 2D Fourier transform of $\mathcal{C}$ (by the Wiener–Khinchin theorem, the power spectral density of ξ normalized by h^2):

$$W(k_x, k_y) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathcal{C}(u, v) \exp\left\{-j\left(k_x u + k_y v\right)\right\} du\, dv, \tag{17}$$

  where $u = x - x'$ and $v = y - y'$ are the lags (separations) in x and y, so that $r = \sqrt{u^2 + v^2}$ in Eq. (14), and

$$k_x = k\left(\sin\theta_s \cos\phi_s - \sin\theta_i\right) \qquad \text{and} \qquad k_y = k \sin\theta_s \sin\phi_s, \tag{18}$$

  with the incident wave in the x–z plane propagating toward $+x$, and where

  - $\theta_s \equiv$ scattering angle;
  - $\phi_s \equiv$ scattering azimuth (the angle between the plane of incidence and the propagation vector of the scattered wave, measured in the horizontal plane from the forward, $+x$, direction).

- Ex.: for the case where a signal is transmitted and received by the same antenna, $\theta_s = \theta_i$ and $\phi_s = \pi$ $\Rightarrow (k_x, k_y) = (-2k \sin\theta_i, 0)$.

- For the common correlation functions, we have

  - Gaussian:

$$\boxed{W(k_x, k_y) = \frac{\ell^2}{2} \exp\left\{\frac{-\left(k_x^2 + k_y^2\right)\ell^2}{4}\right\}} \tag{19}$$

  - Exponential:

$$W(k_x, k_y) = \frac{\ell^2}{\left[1 + \left(k_x^2 + k_y^2\right)\ell^2\right]^{3/2}} \tag{20}$$

3 Scattering from an undulating surface

- Commonly called the Kirchhoff or physical optics model.

- Applicable to surfaces with gentle undulations whose wavelength $\gg$ EM wavelength. This allows us to

  1. approximate the surface as a series of infinite planes that are tangent to the surface (the assumption of infinite planes allows us to neglect boundary effects);
  2. use the concepts of reflection and transmission at a planar boundary, as discussed in previous lectures (Lectures 4–6).

- The details get messy, and we won't go through them in this course (I'm happy to provide references for those who are interested).

3.1 Basic concepts

See Figure 3.

- We can calculate the scattered and transmitted fields just as we did for the case of a planar interface (because all we've done is approximate the undulating surface as a bunch of planar surfaces)

  $\Rightarrow$ the Fresnel reflection and transmission coefficients (Lecture 5) will play a role.

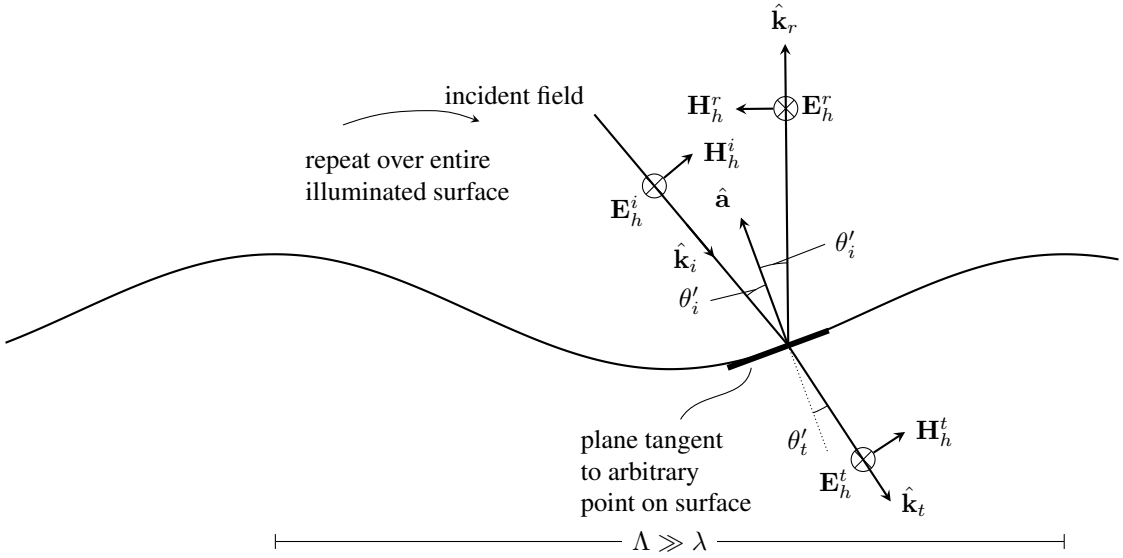


Figure 3: Kirchhoff (physical optics) approximation: at each point of a gently undulating surface, the incident wave is reflected and transmitted as at the infinite plane tangent to the surface, with local angles θ_i' and θ_t' measured from the local normal $\hat{\mathbf{a}}$. H polarization is shown.

3.2 Backscatter toward the source

- To keep things simple and the course moving forward, we'll only consider the question of how much power is scattered back toward the source (as is the general case in "active" remote sensing).
- Again, skipping the details and simply noting that to do so, we need to consider the entire scattered field $\Rightarrow$ integrate over the surface

  $\rightarrow$ done by application of Green's theorem.
- To get an analytical solution, it is necessary to make an assumption about the RMS height of the surface, h.

3.2.1 Limit of small h, so that the surface slopes are shallow

- Allows for a reduced model for backscatter, commonly called the "scalar approximation" (pretty terrible name, but that's what we have).
- The "scalar approximation" expression for backscatter is pretty gnarly, and we won't repeat it here $\rightarrow$ see Ulaby et al. (1986), Ch. 12, for the expression and details of the derivation.
- Key point is that shallow slopes allow for both coherent (specular) and incoherent (diffuse) scatter $\rightarrow$ in general, backscattered power will increase with roughness height (i.e., steepening surface slope) and with decreasing wavelength.

3.2.2 Limit of large h ($h \equiv$ RMS roughness height)

- Allows us to assume the gradient of the phase is negligible ("stationary phase approximation").

- It can be shown in this case (Gaussian correlation function) that

$$\sigma_{pp}^0 = \frac{|\rho_p(0)|^2}{4m_s^2 \cos^4\theta_i} \exp\left\{-\frac{\tan^2\theta_i}{4m_s^2}\right\}, \tag{21}$$

  where

  - $p = h$ or v polarization;
  - $m_s = h/\ell \equiv$ characteristic surface slope;
  - $\rho_p(0) = \rho(0) \equiv$ Fresnel reflection coefficient at normal incidence (the Fresnel coefficient evaluated at $\theta_i = 0$; not the correlation function of Eq. (14), for which $\mathcal{C}(0) = 1$).

  (Note: $\sigma_{hv}^0 = \sigma_{vh}^0 = 0$ in this model $\Rightarrow$ no depolarization.)

- Plotting this expression for $\theta_i = 45^\circ$ and a range of slopes gives Figure 4.

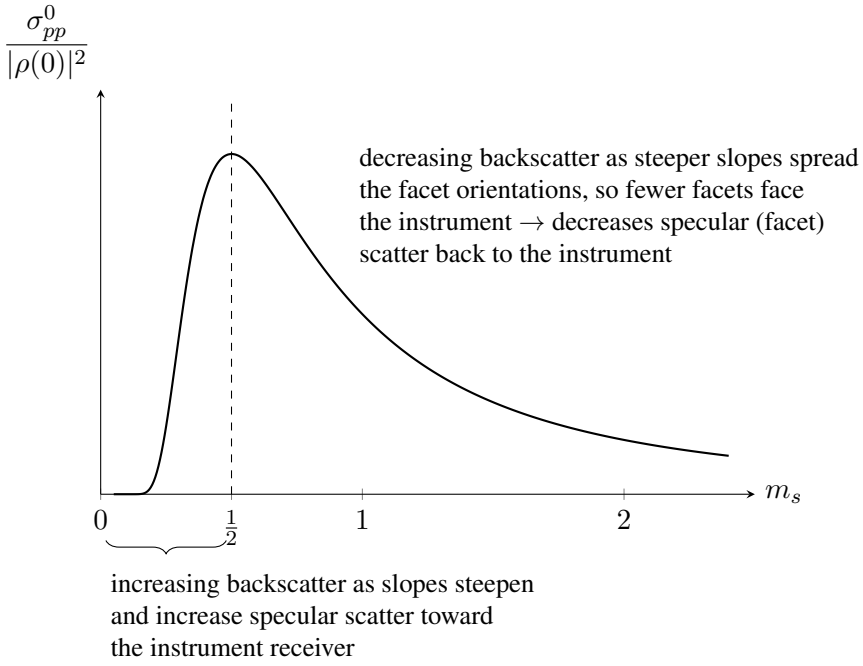


Figure 4: Backscattering coefficient of Eq. (21), normalized by $|\rho(0)|^2$, versus the characteristic surface slope m_s for $\theta_i = 45^\circ$.

- The applicability of the model to natural terrain is limited due to the restrictions of the assumptions, but the intuition we build from the approach (simplifying assumptions) and the results (backscatter power is amplified for certain terrain) is valuable for more realistic scattering models.

Variables

Units are SI. Values are given for universal constants; "typical" values are representative, not definitive.

Symbol	Description	Units	Value
Constants and wavenumbers			
j	imaginary unit, $j^2 = -1$ (time dependence $e^{j\omega t}$)	–	–
k	angular wavenumber in the medium above the surface	rad/m	$k_0 = 2\pi/\lambda$ for air
λ	wavelength (radar)	m	–
d_{ant}	largest dimension of the antenna (source)	m	far field: $R \geq 2d_{\text{ant}}^2/\lambda$
Radar equation			
R	distance (range); in the far field $R \gg \lambda$	m	–
$R_t,\ R_r$	distances from the transmit antenna to the scatterer and from the scatterer to the receiver	m	$R_t = R_r = R$ (same antenna)
P_t	transmit power	W	–
$G_t,\ G_r,\ G$	gains of the transmit and receive antennas; common gain	dimensionless	$G_r = G_t = G$ (same antenna)
$\mathcal{S}_s$	power density at the scatterer (magnitude of the Poynting vector)	W/m^2	$P_t G_t/(4\pi R_t^2)$
$A_{\text{eff},s}$	effective receiving area of the scatterer	m^2	unknown
P_{rs}	total power received by the scatterer	W	$\mathcal{S}_s A_{\text{eff},s}$
f_a	fraction of power absorbed by the scatterer	dimensionless	0–1
P_{ts}	power radiated by the scatterer	W	$P_{rs}(1 - f_a)$
G_{ts}	gain of the scatterer in the direction of the receiver	dimensionless	unknown
$\mathcal{S}_r$	power density at the receiver	W/m^2	–
A_{eff}	effective area of the receiving aperture	m^2	$\lambda^2 G_r/(4\pi)$ (lossless antenna)
P_r	power entering the receiver	W	–
σ	radar (scattering) cross section, $A_{\text{eff},s}(1 - f_a)G_{ts}$	m^2	–
$\sigma_{(n)},\ A_{(n)}$	radar cross section of individual scatterer n and the horizontal (ground-projected) area illuminated around it	m^2	–
σ^0	normalized radar cross section ("sigma-naught"), $\langle \sigma_{(n)}/A_{(n)} \rangle_n$	dimensionless (m^2/m^2)	–
$p,\ q$	polarization of the received and transmitted waves	–	h or v
$P_p^r,\ P_q^t$	p-polarized received power; q-polarized transmit power	W	–
σ_{pq}^0	normalized radar cross section for receive/transmit polarizations p, q	dimensionless	$\sigma_{hv}^0 = 0$ in Eq. (21)

Symbol	Description	Units	Value
A	nominal area of illumination: horizontal (ground-projected) area illuminated, set by the radar geometry	m^2	–
$E_p^s,\ E_q^i$	scattered (p-polarized) and incident (q-polarized) electric fields	V/m	–
Γ_{pq}	reflectance, $\lvert E_p^s\rvert^2/\lvert E_q^i\rvert^2$ (depends on range, $\propto 1/R^2$)	dimensionless	–
Surface scattering			
$x,\ y,\ z$	Cartesian coordinates	m	surface in the x–y plane
$\hat{\mathbf{k}}_i,\ \hat{\mathbf{k}}_r,\ \hat{\mathbf{k}}_t$	propagation unit vectors of the incident, reflected, and transmitted waves	dimensionless	–
θ_i	angle of incidence	rad	–
Surface roughness			
$\xi(x, y)$	local surface height	m	zero mean
h	RMS height, $\langle \xi^2 \rangle^{1/2}$	m	–
$x',\ y'$	coordinates of a second point on the surface	m	–
$\mathcal{C}(r)$	surface-height auto-correlation function	dimensionless	$\mathcal{C}(0) = 1; \mathcal{C}(\ell) = e^{-1}$
r	horizontal separation between two surface points	m	–
ρ	Fresnel reflection coefficient of a planar interface (Lectures 4–5)	dimensionless	–
Γ	reflectivity of an infinite planar interface, $\lvert\rho\rvert^2$	dimensionless	0–1
ℓ	correlation length	m	–
W	roughness spectrum: 2D Fourier transform of $\mathcal{C}$ (normalized height power spectrum)	m^2	–
$k_x,\ k_y$	spectral wavenumbers of the surface roughness	rad/m	backscatter: $(-2k\sin\theta_i, 0)$
$u,\ v$	separations in x and y (lags)	m	–
θ_s	scattering angle	rad	θ_i in backscatter
ϕ_s	scattering azimuth	rad	π in backscatter
Undulating surface			
Λ	wavelength of the surface undulations	m	$\Lambda \gg \lambda$
$\hat{\mathbf{a}}$	local unit normal to the surface	dimensionless	–
$\theta_i',\ \theta_t'$	local angles of incidence and transmission (from $\hat{\mathbf{a}}$)	rad	–
$\mathbf{E}_h^i,\ \mathbf{H}_h^i,\ \ldots$	H-polarized incident, reflected, and transmitted fields	V/m; A/m	–
m_s	characteristic surface slope, h/ℓ (Ulaby & Long's rms slope is $\sqrt{2}\,h/\ell$)	dimensionless	peak backscatter at $m_s = 1/2$ for $\theta_i = 45°$
$\rho_p(0)$	Fresnel reflection coefficient at normal incidence (p polarization)	dimensionless	–
σ_{pp}^0	co-polarized normalized radar cross section	dimensionless	–